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DEPARTMENT : B.E CSE

SUBJECT : PYTHON

BATCH: 5

Mini Project 5: Swap Two Numbers Without Third Variable in Python

1. Project Title

Swap Two Numbers Without Using a Third Variable in Python

2. Aim

To design a Python program that allows the user to input two numbers and swap their values without using any extra variable, displaying the result clearly.

3. Problem Statement

To create a Python program that swaps the values of two numbers using arithmetic operators and displays the result. This demonstrates the concept of operator-based manipulation in Python.

4. Objectives

- To swap two numbers without using a third variable.
- To use arithmetic operators effectively.
- To strengthen logical thinking in programming.
- To understand basic input/output handling in Python.

5. Tools Used

- Python 3
- VS Code / IDLE

6. Description

This project allows the user to enter two numbers and swaps their values using arithmetic operators. The program avoids the need for a temporary variable, making it more efficient and demonstrating a classic programming trick.

7. Algorithm

1. Start
2. Read two numbers (a and b) from the user
3. Swap the numbers using arithmetic operators:
 - $a = a + b$
 - $b = a - b$
 - $a = a - b$
4. Display the swapped values
5. Stop

8. Program (Python Code)

```
# Swap two numbers without using a third variable
```

```
# Input from user
```

```
a = int(input("Enter first number: "))
```

```
b = int(input("Enter second number: "))
```

```
# Swapping using arithmetic operators
```

```
a = a + b
```

```
b = a - b
```

```
a = a - b
```

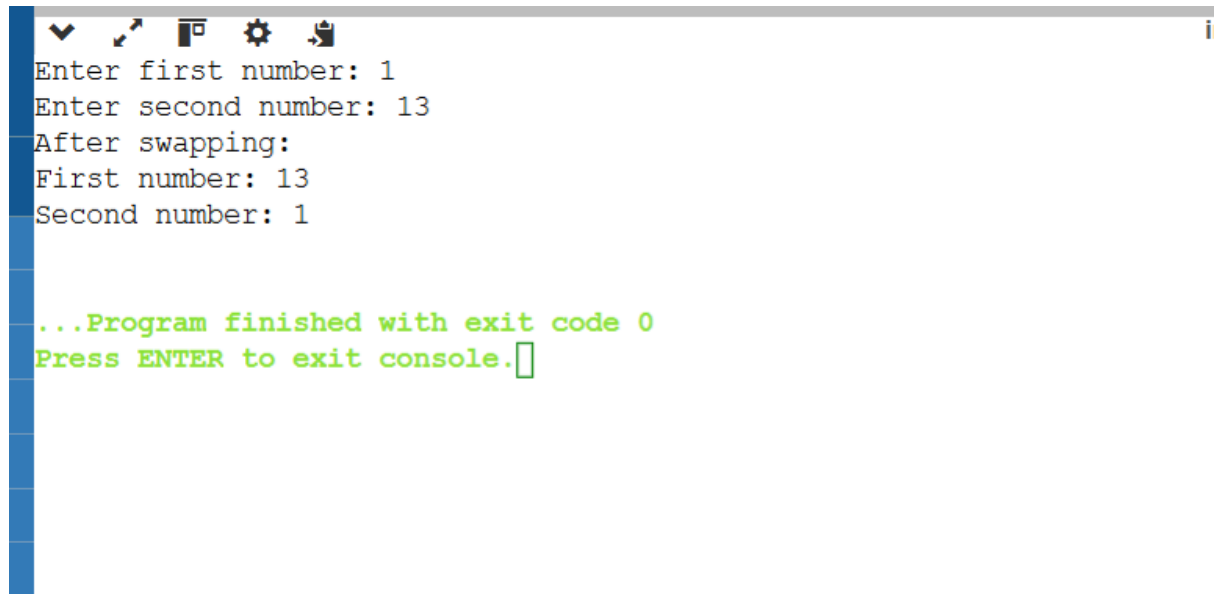
```
# Display the swapped values
```

```
print("After swapping:")
```

```
print("First number:", a)
```

```
print("Second number:", b)
```

9. Output

A screenshot of a terminal window with a dark background and light blue text. The terminal shows the execution of a Python program. The user enters '1' for the first number and '13' for the second number. The program outputs 'After swapping:', 'First number: 13', and 'Second number: 1'. At the bottom, it shows '...Program finished with exit code 0' and 'Press ENTER to exit console.' with a cursor. The terminal has a standard toolbar at the top with icons for window management and settings.

```
Enter first number: 1
Enter second number: 13
After swapping:
First number: 13
Second number: 1

...Program finished with exit code 0
Press ENTER to exit console.
```

10. Result

The program successfully swaps the values of two numbers without using any extra variable and displays the swapped result.

11. Conclusion

This project demonstrates a classic method of swapping numbers using arithmetic operators. It reinforces understanding of operators, input/output handling, and logical problem solving in Python.